Final Project 2 — Hand-Gesture Recognition for Mobile Robot Control

1. Project Overview

In industrial environments, mobile robots increasingly operate alongside human workers. Effective human–robot interaction requires intuitive, reliable, and efficient communication methods.

Hand-gesture recognition provides an alternative communication method, allowing operators to control robots through simple visual commands without physical controllers.

In this project, your group will develop a vision-based hand-gesture recognition system and integrate it with a mobile robot in a simulated environment.

The system should recognize hand gestures from RGB images, translate them into predefined robot commands, and control a mobile robot to perform basic movements such as moving forward, turning left, turning right, and stopping.

2. Required Tasks

Task 1 — Literature Review and Gesture Command Design

Conduct literature research on hand-gesture recognition techniques and their applications in human–robot interaction.

Your group must design a command vocabulary with at least four gesture-based robot commands, plus NO COMMAND. The dataset labels are not robot commands: your group selects the gestures and explains why they are suitable for robot control. NO COMMAND is the output when the hand shows none of your command gestures (for example, a no_gesture hand or a gesture you did not select) or when the prediction is uncertain.

When designing the vocabulary, consider:

- Are the chosen gestures visually distinguishable and intuitive for a human operator?
- Which gesture pairs may be easily confused?
- Which commands could be dangerous if triggered incorrectly, and how should the system behave when the gesture is uncertain?

Required output: A literature review, a clearly defined gesture-to-command mapping, and justification of the selected recognition approach.

Task 2 — Dataset Preparation and Generalization Evaluation

All groups must use the HaGRID Sample 30K dataset for model development and evaluation.

Download hagrid-sample-30k-384p.zip (about 1 GB) from the "Files and versions" tab of the dataset page. The hand boxes, labels and user_id are in the JSON files inside the zip; the online viewer does not show them. Cite: A. Kapitanov, K. Kvanchiani, A. Nagaev, R. Kraynov and A. Makhliarchuk, "HaGRID – HAnd Gesture Recognition Image Dataset", WACV 2024. Licence: the HaGRID licence, based on CC BY-SA 4.0.

A reliable gesture recognition system should work on people who were not included in model training. You must therefore prepare two evaluation protocols:

Protocol A — Random Image Split: randomly divide the images into training, validation, and testing subsets; images of the same person may appear in multiple subsets.

Protocol B — Subject-Independent Split: divide the dataset according to user_id, so that no person appears in more than one subset.

Required activities:

- Prepare reproducible splits for both protocols with reasonable class balance, and report the number of users and images in each split.
- Analyze the data distribution and the differences between people, backgrounds, hand appearances, and lighting conditions.
- Train and evaluate the recognition methods under both protocols, and analyze whether random splitting overestimates performance on unseen people.

Task 3 — Gesture Recognition and Reliable Command Generation

Implement a vision system that takes a single RGB image as input and predicts one of the predefined robot commands or NO COMMAND. You may choose your own techniques, for example HOG + SVM, a small CNN, or hand landmarks from a pretrained model (e.g., MediaPipe Hands) with a classifier. Implement at least two approaches or clearly differentiated model configurations for comparison.

In robotic applications, high classification accuracy alone is insufficient. A missed FORWARD gesture only causes a delay, but a resting hand recognized as FORWARD causes unintended motion, and STOP confused with FORWARD is hazardous. Your group must therefore develop a command validation or rejection mechanism.

Required activities:

- Implement hand localization and image preprocessing. The provided hand boxes may be used to crop hands for training; at test time, the system must find the hand itself.
- Implement and evaluate at least two recognition approaches, producing gesture predictions with confidence scores or another reliability measure.
- Define when the system rejects a prediction, and establish a method to reduce false robot commands.
- Investigate how the rejection mechanism affects recognition accuracy and missed commands.

Task 4 — Integration with Mobile Robot Control

Extend the gesture recognition system to control a simulated differential-drive mobile robot; advanced navigation, SLAM, and path planning are not required. You may use a basic Python 2D simulator, PyBullet (on Windows or macOS, install it from conda-forge), ROS/ROS 2 with Gazebo, or another suitable platform.

Required activities:

- Build a simple mobile robot simulation whose motion is driven by the actual output of the vision system, not by manually chosen gesture labels.
- Demonstrate at least four commands plus NO COMMAND in a complete operation, for example driving through a simple virtual workshop with a predefined gesture sequence.
- Ensure that an uncertain or rejected prediction does not trigger unintended movement: NO COMMAND produces no new movement, and STOP immediately cancels the current movement.

Task 5 — Quantitative Evaluation and Comparative Analysis

Evaluate the complete system from gesture recognition to robot control. Your experiments must address the following four aspects:

- Gesture recognition accuracy
- Generalization to unseen people
- False-command rate: how often the system generates a movement command when no command is intended
- Processing speed on a CPU: time per image, from the input photo to the robot command

3. Final Presentation and Deliverables

The final group presentation should include:

- Problem definition and application scenario — Why is hand-gesture-based robot control useful in industrial environments?
- Literature review — What gesture recognition approaches have been investigated?
- Gesture and command design — Which gestures are selected, and how are they mapped to robot commands?
- Dataset preparation — How are the dataset and subject-independent splits prepared?
- Recognition methodology — How do the selected algorithms recognize gestures?
- Command reliability mechanism — How does the system avoid unintended commands?
- Mobile robot integration — How are recognized gestures used to control the simulated robot?
- Experimental evaluation — What are the recognition accuracy, generalization performance, false-command rate, and CPU processing speed?
- Comparison and failure-case analysis — Which method is more suitable, and why?
- Conclusion — What engineering insights were obtained?

4. Bonus Tasks — Additional Marks

Propose a meaningful enhancement to one of your recognition methods or to the command reliability mechanism. The improvement should be evaluated against your own baseline to demonstrate its effectiveness.